

Upgrading Ethanol to Higher Alcohols via Biomass-Derived Ni/Bio-Apatite

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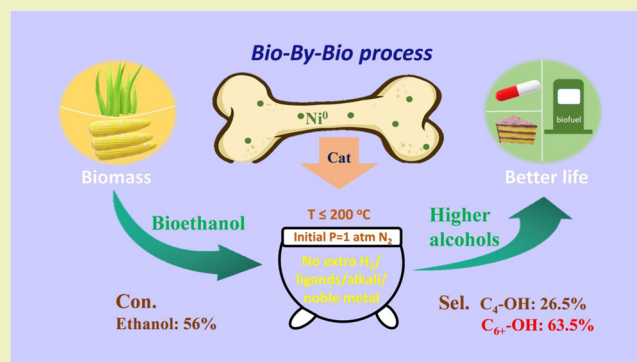
ABSTRACT: Acquiring value-added chemicals from renewable ethanol instead of fossil resources has special significance under the background of carbon neutrality. In this work, a heterogeneous recyclable biomass-derived Ni/bio-apatite catalyst was developed for upgrading ethanol to higher alcohols (C_{6+} -OH). Catalysts were prepared employing calcined porous natural bone and analyzed by various characterizations of thermogravimetric analysis–differential thermal analysis, X-ray diffraction, high-angle annular dark field scanning transmission electron microscopy, X-ray photoelectron spectroscopy, H_2 -temperature-programmed reduction, and CO_2 -temperature-programmed desorption. The ethanol upgrading reaction can be achieved in the liquid phase without alkali additives, ligands, and extra hydrogen. The selectivity for C_{6+} -OH reached as high as 67.7% at the single pass 55.6% ethanol conversion, substantially higher than the Anderson–Schulz–Flory distribution. Research shows that the porous structure and coordination between metal and alkaline sites could play key roles in C_{6+} -OH selectivity. The catalyst recycles and reaction pathway of ethanol upgrading to higher alcohols were also discussed.

KEYWORDS: higher alcohol, clean fuel additives, biomass hydroxyapatite, bioethanol, Guerbet reaction

■ INTRODUCTION

Bioethanol has been increasingly used as an oxygenated additive in fuel in recent years.^{1–3} However, its poor compatibility, high hygroscopicity, low flash point, relatively high volatility, and low energy density limit its application in fuel with a high content. Better fuel properties can be achieved by substituting ethanol with C₆₊-OH. It has been demonstrated as a promising bioderived drop-in fuel without modification of vehicles.⁴ In addition, C₆₊-OH has also been widely applied in food, drinks, cosmetics, perfumes, drugs, textiles, plasticizers, detergents, insecticides, coatings, and so forth.^{5,6} However, to date, efficient approaches for acquiring C₆₊-OH using bioderived materials as feedstocks remain limited.

Upgrading from bioethanol may satisfy the abovementioned goals, in which the catalyst with high performance plays a key role. Pioneering scientists explored some good catalytic systems. Wass and co-workers⁷ reported a Ru-based homogeneous catalyst with a ligand of *bis*(diphenylphosphino)methane for ethanol upgrading, showing an ethanol conversion of 22% and butanol selectivity of over 93%. Szymczak and co-workers⁸ synthesized amide-derived *N,N,N*-Ru(II) complexes to upgrade ethanol to butanol, increasing the conversion to 53%. Liu⁹ developed a [Mn] complex using manganese as an alternative to noble metals, achieving an ethanol conversion of



28.6% and butanol selectivity of 87%, significantly reducing the cost of the catalyst. Although homogeneous catalysis shows remarkable efficiency, C₆₊-OH selectivity is extremely low.^{7–13} Some heterogeneous catalysts, such as zeolite catalysts,¹⁴ metal oxide^{15–18} and hydroxyapatite (HAP),^{19–22} have also been used to catalyze this reaction. However, the reaction conditions are often very harsh ($T > 350\text{ }^{\circ}\text{C}$), resulting in C₆₊-OH selectivity being sacrificed for gas byproducts. The addition of transition metals such as Ir, Pd, Ru, Co, Cu, or Ni can reduce the reaction temperature.^{23–26} However, their focus was on the condensation of ethanol to butanol instead of adjusting the active sites of the catalyst to improve the selectivity of C₆₊-OH.

HAP is attractive because it has strong basic sites that can promote aldol condensation. HAP is one of the main components of bones. It can be obtained by calcination to remove organic matter such as protein and cellulose.^{27,28} In fact, these organic matter materials are similar to natural templating agents, and removing them can form mesoporous

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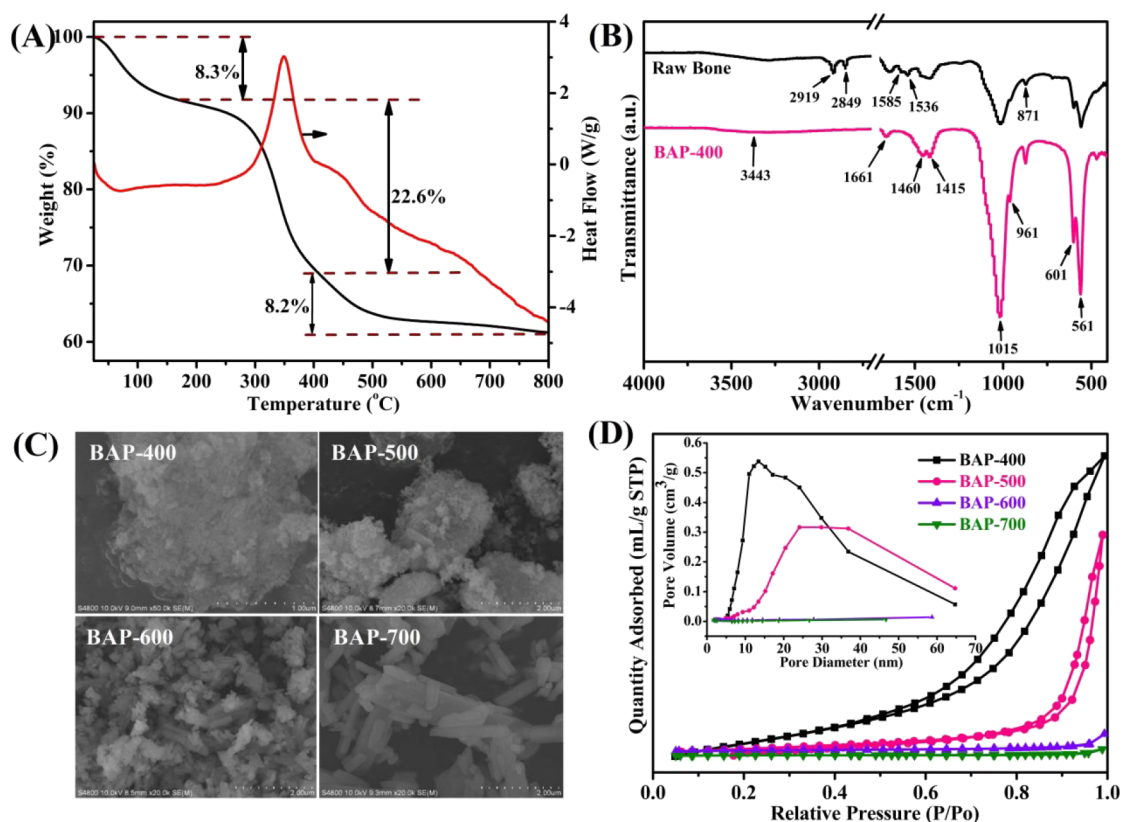


Figure 1. (A) TG and DSC plots of the pork bone; (B) FTIR spectrum of the raw bone and BAP-400; (C) SEM images of BAP calcined at 400, 500, 600, and 700 °C; and (D) nitrogen sorption isotherms and pore size distribution of BAP calcined at 400, 500, 600, and 700 °C.

HAP. Therefore, obtaining HAP from bones is an economically viable and environmentally friendly choice. Among many metals, Ni is favored because of its high activity and low cost.²⁹ However, Ni can also cause the rupture of carbon–carbon bonds, which increases side reactions and prohibits the formation of higher alcohols. In this work, to obtain high selectivity of C_{6+} -OH from bioethanol, a high efficiency catalyst was prepared by loading Ni over biomass (pork bone)-derived apatite. Special attention has been paid to the influence of the pore structure and catalytic site coordination on the reaction results. The catalyst regeneration and reaction pathway of ethanol upgrading to higher alcohols over the Ni-BAP catalyst are also discussed.

EXPERIMENTAL PROCEDURES

Materials. Nickel nitrate hexahydrate ($Ni(NO_3)_2 \cdot 6H_2O$, 99% purity) and ethanol (99.5% purity) were purchased from Sigma-Aldrich. Pork bones were obtained agreeing with biological ethics because they were waste bones originally discarded from the supermarket. Deionized water was obtained from a water purification system (Millipore, Direct-Q3). Reagents including toluene, tetrahydrofuran, pentane, *n*-butanol, *n*-hexanol, 2-ethyl-1-butanol, *n*-octanol, and 2-ethyl-1-hexanol were purchased from Adamas-beta with analytical purity and used without further purification.

Catalyst Preparation. The bio-apatite used as support in the experiments was obtained from pork bones. The preparation process was as follows: crush the bone and wash with boiling water to remove the dirt and protein on the surface. Then, the bone fragments were dried and calcined in a muffle furnace at 300–700 °C for 4 h in an air atmosphere to remove organic matter. The calcined solids were ground to fine powders and denoted as BAP-*T*, where *T* is the calcination temperature. Ni was loaded on BAP-*T* by the ethanol-assisted impregnation method followed by H_2 reduction. Typically, a

certain quantity of $Ni(NO_3)_2 \cdot 6H_2O$ was dissolved in 40 mL of ethanol, and 2 g of BAP-*T* powders was added with agitation overnight. Then, solvents were removed by rotary evaporation in vacuum. The dried powders were reduced in a H_2 flow for 2 h at 470 °C and then stored in a nitrogen-filled glovebox, labeled BAP-*T*-aNi. For example, the BAP-400-SNi sample means that a 5% mass fraction of Ni is supported on BAP powders calcined at 400 °C. (For convenience, without special instructions, BAP in the following refers to samples calcined at 400 °C).

Catalyst Characterization. The pork bone was subjected to thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) using a METTLER TOLEDO Simultaneous Thermal Analyzer. A total of 5 mg of the bone sample was heated in an alumina crucible at a rate of 10 °C/min from 25 to 800 °C in air.

Fourier transform infrared spectroscopy (FTIR) measurements were carried out in ATR mode with a resolution of 4 cm^{-1} and a range of 400–4000 cm^{-1} by using a NICOLET iS50 FT-IR.

X-ray diffraction (XRD) analysis was performed using a Rigaku SmartLab 3KW instrument with Cu K α radiation (40 kV, 30 mA).

The morphologies of the samples were analyzed by scanning electron microscopy (SEM) using a Hitachi S-4800 scanning electron microscope.

Transmission electron microscopy with energy-dispersive X-ray spectroscopy (TEM/EDS) and high-angle annular dark field scanning transmission electron microscopy (HAADF-STEM) were performed on an FEI Tecnai F20 transmission electron microscope.

X-ray photoelectron spectroscopy (XPS) data were collected using an XPS system (Thermo Fisher Scientific, Thermo K-Alpha+) equipped with a microfocused, monochromatic Al K α X-ray source (1486.6 eV).

Nitrogen adsorption–desorption isotherms were measured with a Micromeritics ASAP 2460 system at 77 K. Prior to the nitrogen sorption analysis, all samples were heated to 300 °C under vacuum for 1 h to remove moisture and adsorbed gases. The Brunauer–Emmett–Teller (BET) method was utilized to calculate the catalyst surface

area. The pore size distribution curve was obtained from the desorption branch by using the Barrett–Joyner–Halenda (BJH) method.

H₂-TPR was carried out with a TP-5080 instrument equipped with a thermal conductivity detector (TCD). The sample was first pretreated under a He atmosphere at 300 °C for 1 h. After cooling to 30 °C, the carrier gas was switched to 5% H₂/N₂ (v/v) at a total flow of 50 mL/min. When the baseline was stable, the sample was heated to 800 °C at a heating rate of 10 °C/min, and the amount of H₂ consumed was measured by the TCD.

The CO₂ temperature-programmed desorption (CO₂-TPD) experiment was carried out with a TP-5080 instrument equipped with a TCD. A total of 60 mg of the freshly reduced catalyst was pretreated under a He atmosphere at 300 °C for 1 h. After cooling to 100 °C, CO₂ (30 mL/min) was introduced for 30 min. Then, the CO₂ was switched to He purge gas (30 mL/min) to sweep away physically adsorbed CO₂ on the surface of the catalyst. The TPD measurements were conducted from 100 to 600 °C at a heating rate of 10 °C/min in a He flow (30 mL/min). The desorbed CO₂ was measured by a TCD.

Catalytic Performance Evaluation. Reactions were carried out in a PTFE-lined stainless steel autoclave. Typically, 0.5 mL of ethanol and 0.05 mL of tetrahydrofuran as the internal standard, 2.5 mL of toluene, and 0.25 g of the catalyst were added to a 25 mL autoclave purged with N₂ flow. Initially, the pressure was 0.1 MPa before heating. However, the autogeneous pressure of ca. 1 MPa is generated at reaction temperature (200 °C). After the reaction, the autoclave was quickly cooled in icy water, and then the product composition was analyzed by GC-FID (Agilent 8890 GC System equipped with an HP-5 capillary column). The gaseous products were analyzed using an Agilent 7890B GC (TCD) system equipped with an HP-5 column.

For scale-up experiments, the reactions were conducted in a 100 mL autoclave (Parr 100 mL) to achieve better accuracy and stability. Material balance was also evaluated, and the mass loss was less than 5%.

The conversion of ethanol and the selectivity of higher alcohols are calculated as follows:

$$\text{conversion (\%)} = \frac{\text{ethanol reacted}}{\text{ethanol added}} \times 100\% \quad (1)$$

$$\text{selectivity (\%)} = \frac{\text{ethanol consumed to produce the target product}}{\text{ethanol reacted}} \times 100\% \quad (2)$$

RESULTS AND DISCUSSION

Catalyst Textural and Physicochemical Properties.

TGA–differential thermal analysis (DTA) curves for the heat treatment of bone are shown in Figure 1A. Three successive steps of weight loss can be observed in the TG curve. The first weight loss of ca. 8 wt % occurred at 80–185 °C should be attributed to the volatilization of water and volatile organic matter. The second weight loss of 22.6% is observed from 281 to 402 °C due to the oxidation of organic matter in the natural inorganic–organic hybrid network, as we can correspondingly find a strong exothermic peak in the DTA curve. Heat treatment above 400 °C could cause further mineralization and crystal growth accompanied by the release of CO₂ due to carbonate decomposition.^{30,31} These results suggested that calcination at 400 °C can remove most of the organic matter to obtain bio-apatite.

The morphology and textural structure of bio-apatite were studied during heat treatment. As the calcination temperature increased, the material started mineralization since 400 °C, and the nanoparticles ultimately grew into larger perfect nanocrystals (Figure 1C) with the disappearance of the pore structure above 600 °C (Figure 1D). As shown in Figure 1D, the isotherms of bio-apatite samples calcinated at 400 and 500

°C showed type IV isotherms with a H3 hysteresis loop,^{32,33} indicating the presence of abundant mesopores, which may lead to a different catalytic performance.^{34–36} The pore size distribution moved from ca. 5–50 to 10–70 nm when the temperature rose from 400 to 500 °C, as calculated by the BJH method. The samples calcinated at 600 and 700 °C exhibited nearly nonporous structures with low adsorption capacity. The specific surface area dramatically decreased from 90 m²/g (BAP-400) to 3 m²/g (BAP-700).

Surface functional groups were studied by FTIR analysis. As shown in Figure 1B, the peaks at 2919, 2849, 1585, and 1536 cm^{−1} in the spectra of raw bone are assignable to the stretching vibrations of C–H and N–H,³⁷ while they disappeared in BAP-400, indicating that most of the organic matter can be removed after calcination at 400 °C, which is consistent with the TGA results. The spectra of the two samples exhibit absorption bands at 561, 601, 961, and 1015 cm^{−1}, which are assignable to phosphate ions PO₄^{3−}.^{38,39} These bands are relatively weak in the raw bone due to the cross-linking structure of the organics and inorganics, while they are significantly enhanced in BAP-400 due to mineralization.⁴⁰ In addition, the peaks at 1460, 1415, and 871 cm^{−1} belong to the CO₃^{2−} functional group in BAP.^{38,39}

Figure 2 shows the XRD of the typical catalysts. BAP-5Ni and BAP-0.25Ni showed an apatite-like crystalline form.

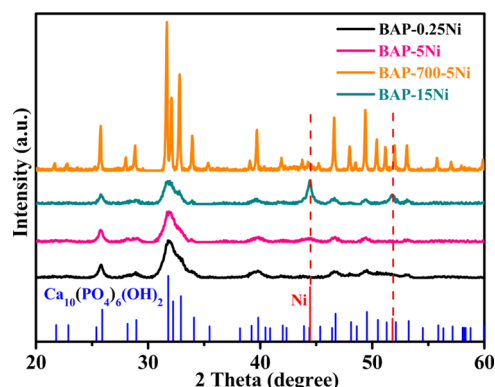


Figure 2. XRD patterns of different catalysts.

However, compared with the standard HAP pattern (PDF card #74-0566), the peaks were much broader/indistinct, indicating that the crystal phase of BAP was defective and far from perfect. Regarding BAP-700-5Ni, which experienced heat treatment at 700 °C, the diffraction peaks became clear/sharp and consistent with HAP, suggesting that the crystal phase tends to be perfect. The nickel diffraction peaks at 44.8 and 51.8° shown in the BAP-15Ni sample are assignable to the (111) and (200) planes of Ni-hcp metal (PDF card # 87-0712), while they were not observed for BAP-0.25Ni and unclear for BAP-5Ni, indicating a relatively good dispersion of Ni species at low loading.

The experimental N₂ physisorption isotherms and pore size distributions for the three Ni/BAP samples are shown in Figure 3. The Ni deposition induces an apparent loss of BET surface area from 90.2 m²/g for BAP-0.25Ni to 46.4 m²/g for BAP-15Ni, accompanied by a slightly upshifted pore size distribution. This is mainly due to the blockage of the smaller pores in the BAP caused by excessive nickel species. Pores and surface area may affect the distribution of metallic sites and the

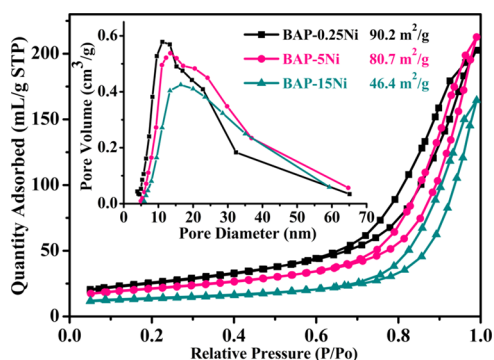


Figure 3. Nitrogen sorption isotherms and pore size distribution of catalysts with different loadings.

number of accessible basic sites, finally leading to changes in catalytic performance and reaction mechanism.

TEM, HAADF-STEM, and EDS images were utilized to clarify the morphology and elemental distribution of the Ni/BAP catalysts. As shown in Figure 4A,B, the BAP consisted of amorphous nanoparticles with obvious mesoporous structures. The Ni particle size showed a normal distribution with an average diameter of 1.4 nm even after a reduction in hydrogen flow (Figure 4C). The quite small Ni particles were well dispersed on the BAP support, exposing a lattice distance of 0.207 nm, which can be assigned to the (111) interplanar spacing of metallic Ni (0.203 nm) (Figure 4D). It is worth noting that Ni particles became larger with increasing Ni loading and decreasing specific surface area of the support (see BAP-5Ni and BAP-700-5Ni, as shown in Figure S1). The elemental mapping of the BAP-0.25Ni catalyst also shows that Ca, P, and O have a similar distribution and that Ni species

were uniformly distributed over the BAP particles (Figure 4E,F).

H₂-TPR tests were performed, and the profiles are shown in Figure 5. The profile of BAP-15Ni shows three distinct peaks

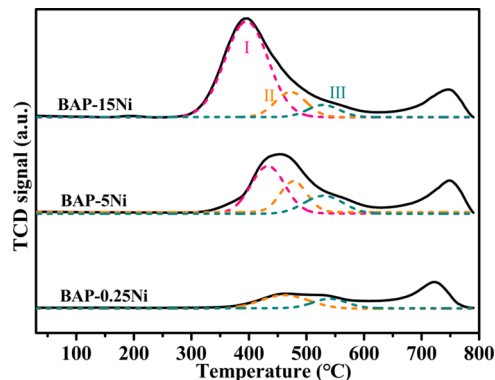


Figure 5. H₂-TPR profiles of several typical catalysts.

centered at 400, 470, and 550 °C before 700 °C. Peak (I) is assigned to the reduction of freestanding NiO particles that have a weak interaction with the support.⁴¹ The second peak is attributed to the reduction of small NiO particles exhibiting a strong interaction with the support.⁴² The third peak ca. 550 °C can be related to the nickel species in the BAP skeleton by ion exchange. With reference to BAP-15Ni, the profile of the BAP-5Ni catalyst shows that the position of the first peak shifts (by 15 °C) toward high temperatures, suggesting a relatively stronger interactions of these species with the BAP support. When the nickel loading was reduced to 0.25 wt %, the peak for the freestanding NiO particles disappeared, and all NiO particles had a strong interaction with the BAP support. The

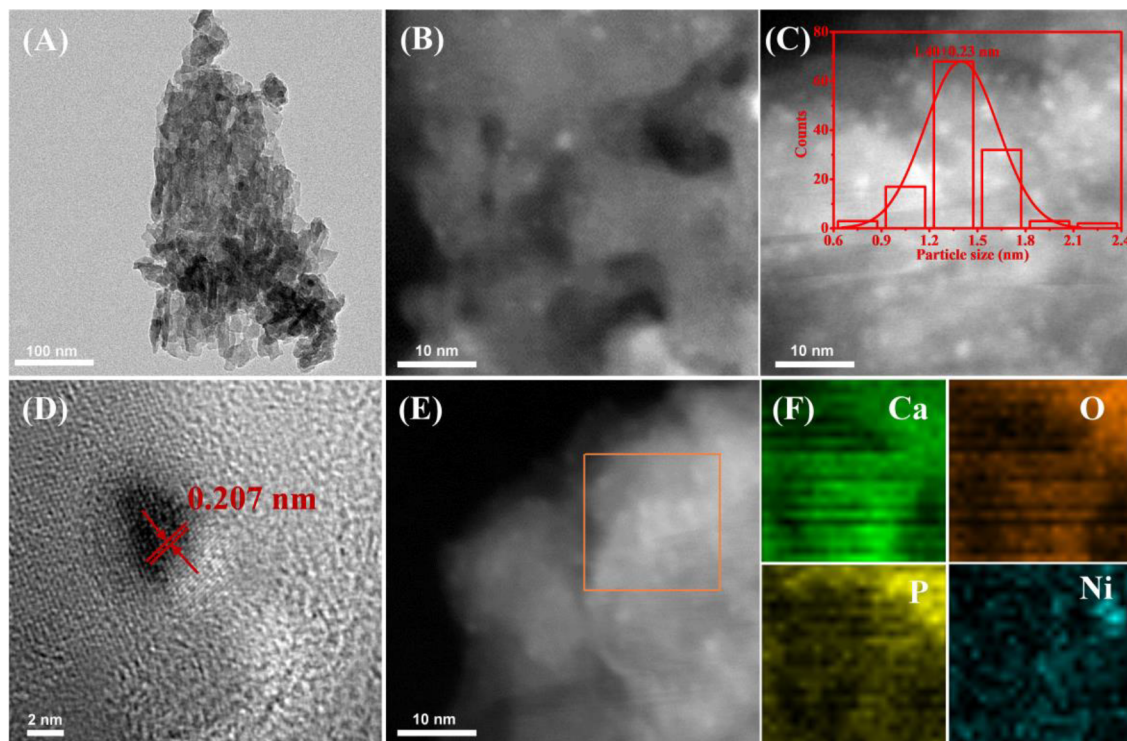


Figure 4. (A–D) TEM and HAADF-STEM images of BAP-0.25Ni; (E, F) EDX elemental mapping of BAP-0.25Ni.

peak at 740 °C on all three samples is attributed to the decomposition of carbonate in the BAP, which can be verified by the H₂-TPR test of a blank BAP support without Ni.

The surface basicity of typical catalysts has been characterized by CO₂-TPD. As shown in Figure 6, three

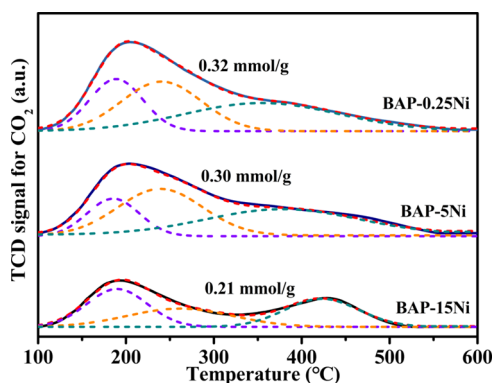


Figure 6. CO₂-TPD profiles of several typical catalysts.

main desorption regions ascribed to weak, medium, and strong basic sites can be observed on the CO₂-TPD diagrams. Compared with BAP-0.25Ni, the number of basic sites in BAP-5Ni is slightly reduced and significantly reduced in BAP-15Ni due to too many Ni species covering the original basic sites. These results showed that the Ni loading amount not only altered the Ni catalytic sites but also changed the distribution of basic sites. Similar to the soluble alkali (NaOH and EtONa) added in a homogeneous reaction system, the basic sites play a key role in aldol condensation. Therefore, the coordination between Ni catalytic sites and basic sites would be significantly influenced, as we will discuss in the pathway section.

To identify the chemical state of the supported nickel, XPS analysis was performed for BAP-0.25Ni and is shown in Figure 7. There are two peaks appearing at binding energies of 853.5

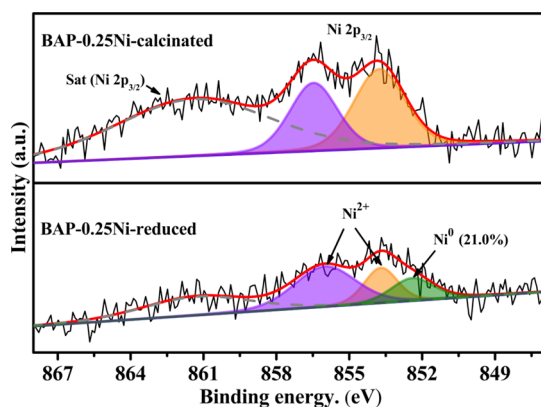


Figure 7. XPS spectra of the Ni 2p_{3/2} region of BAP-0.25Ni.

and 856.4 eV in the XPS spectra of BAP-0.25Ni-calcinated in the Ni 2p_{3/2} region. The fitting peak at 853.5 eV can be ascribed to Ni²⁺ in NiO dispersed on BAP,⁴³ while the peak at 856.4 eV should be attributed to Ni²⁺ in the BAP skeleton due to ion exchange with Ca.⁴⁴ The satellite peak at approximately 861.6 eV is the shakeup satellite peak of nickel at the high binding energy side of the Ni 2p_{3/2} edge.⁴⁵ When the catalyst was reduced at 470 °C for 2 h, the peak at 852.5 eV, which is ascribed to metallic Ni⁰, appeared.⁴⁶ However, the Ni²⁺ species

cannot be completely reduced at this temperature due to the strong interaction between the metal and support.

Catalytic Performance of the Different Catalysts. The effect of calcination temperature on the catalytic performance was studied in the range of 300–700 °C. As shown in Figure 8,

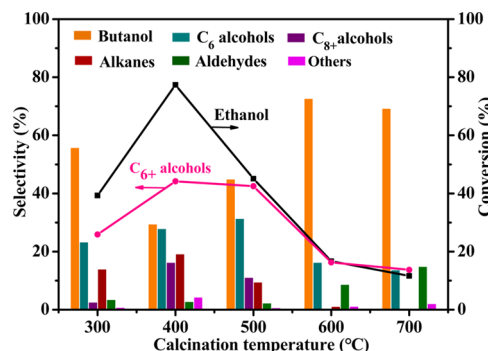


Figure 8. Catalytic performance of the BAP-T-5Ni catalysts ($T = 300, 400, 500, 600,$ and 700). Conditions: 0.1 MPa N₂, 200 °C, 24 h, 2.5 mL of toluene, 0.5 mL of ethanol, 0.25 g of the catalyst, Ni loading 5 wt %.

the conversion of ethanol first rose and reached the optimum with an ethanol conversion of 77.4% at a calcination temperature of 400 °C. Then, the conversion decreased with calcination temperature, showing only 11.6% when the calcination temperature was 700 °C. It is interesting that the product distribution also showed an obvious difference. When the calcination temperature was lower than 400 °C or higher than 600 °C, the primary product was *n*-butanol, whereas the longer chain alcohols, that is, C₆-OH, C₈-OH, and C₁₀-OH, were significantly high at 400 and 500 °C. For example, the selectivity of C₆+ alcohols could reach 44.2% at 400 °C. These results can be explained by the changes in the catalyst structure, as revealed in catalyst characterization. The raw bone was catalytically inert before heat treatment, as we tested the Ni/bone in the reaction and found no effective ethanol conversion. As shown by TG-DTA analysis, the oxidation of organic matter of bone to form the porous structure occurred at ca. 280–400 °C. When the calcination temperature was too low, such as 300 °C, active catalytic centers were hindered by organic matter. However, as revealed by SEM and XRD analyses, when the calcination temperature was too high (above 600 °C), the growth of nanocrystals would significantly reduce the porous structure, leading to a small specific surface area, just as textual analysis showed. Both the accessible metal sites and basic sites are reduced, resulting in weak catalytic activity and low C₆+ alcohols selectivity.

As shown in the catalyst characterization, the Ni loading amount may affect both Ni sites (size, dispersion, and metal–support interaction) and basic sites. Therefore, the Ni loading amount would have a significant influence on the ethanol conversion and distribution of products. As shown in Figure 9, as the Ni loading increased from 0.1 to 5%, the ethanol conversion increased from 17.8 to 77.4%, while the selectivity of total alcohols decreased from 95.2 to 73.7% because some alkanes were generated as byproducts, mainly including odd number chained aliphatic hydrocarbons, that is, CH₄, C₃H₈, C₅H₁₂, C₇H₁₆, and so forth. The selectivity of C₆+ alcohols, including 2-ethyl-1-butanol, 1-hexanol, 2-ethyl-1-hexanol, and 1-octanol, reached a higher level when the Ni loading was in the range of 0.25–1% (Figures S2 and S3 show the typical GC

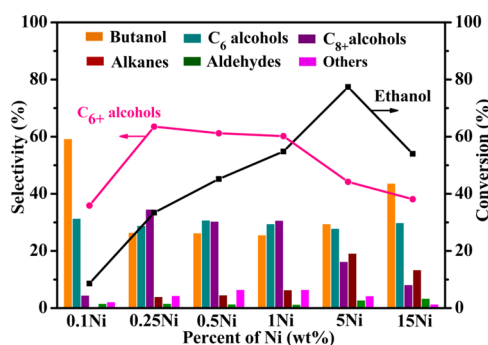


Figure 9. Catalytic performance of the BAP-*a* Ni catalysts (*a* = 0.1, 0.25, 0.5, 1, 5, and 15). Conditions: 0.1 MPa N₂, 200 °C, 24 h, 2.5 mL of toluene, 0.5 mL of ethanol, and 0.25 g of the catalyst.

chromatogram and main products in the liquid and gas phase after ethanol upgrading over BAP-0.25Ni). However, upon further increasing the Ni loading amount to 15%, the ethanol conversion decreased to 54.0%, and low selectivity for C₆+OH was obtained. The optimal Ni loading for ethanol conversion was therefore 0.25 wt %. The BAP-0.25Ni catalyst not only had high selectivity for C₆+OH but also maximized the utilization of Ni; thus, it was used for further investigations. The reasons may be explained as follows: In the conditions of this work, when a low Ni content was loaded, Ni species could be well dispersed over the BAP support with a small cluster size. The dehydrogenation of lower alcohol could be greatly promoted by increasing Ni content, and the aldol reaction is also accelerated with the increase of aldehydes, so the ethanol conversion and C₆+OH selectivity was significantly improved. However, when the amount of Ni was excessive, more BAP support surface was covered, and the basic sites that facilitate the aldol reaction decreased, as evidenced in CO₂-TPD analysis. An increase in the Ni/alkaline site ratio would lead to a low chain propagation trend. Moreover, larger Ni nanoparticles also formed over the support, just as shown by TEM, promoting C–C bond cleavage,²⁴ which caused some aldehydes to convert to alkanes with one carbon less.

Effect of Reaction Temperature. The effect of temperature on ethanol conversion and C₆+OH selectivity was investigated. As shown in Figure 10, when the reaction temperature was increased from 160 to 220 °C, the ethanol conversion increased from 8.3 to 49.3%. The selectivity for C₆+OH reached its highest value at 200 °C. The improved ethanol conversion indicates that ethanol upgrading requires a

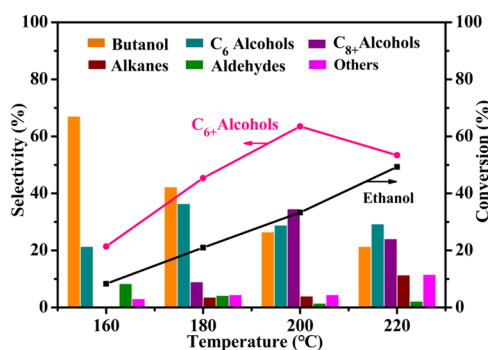


Figure 10. Ethanol was upgraded over the BAP-0.25Ni catalyst at various temperatures. Conditions: 0.1 MPa N₂, 24 h, 2.5 mL of toluene, 0.5 mL of ethanol, and 0.25 g of the catalyst.

relatively higher reaction temperature for its activation. This is consistent with previous studies.^{16,47} In addition, as the temperature increased, the C₄ alcohols/aldehydes were also more reactive, resulting in a higher selectivity for C₆+OH. Upon further increasing the temperature to 220 °C, the overall selectivity for all alcohols decreased to 74% due to the increase in alkanes and other byproducts. This is probably because the activity of aldehyde decarbonylation and self-condensation increased. Based on the reaction efficiency and alcohol selectivity, the reaction temperature was optimized to 200 °C.

Scale-up Experiments. To obtain more reliable data, scale-up experiments were conducted in a 100 mL autoclave under monitoring with continuous sampling. Figure 11 shows the concentrations of ethanol and main products as a function of time. As shown, the porous BAP-0.25Ni and BAP-5Ni catalysts showed high conversion in the scale-up conditions, while the nonporous BAP-700-5Ni catalyst showed low performance. It is worth noting that the concentration variation displayed three different trends. In the case of nonporous BAP-700-5Ni, butanol, C₆-OH and C₈-OH appeared sequentially, agreeing with the sequential propagation reaction mechanism. These results indicated that the Anderson–Schulz–Flory distribution^{48–52} could be proposed in analogy with the case in Fischer–Tropsch synthesis. In the case of BAP-5Ni, butanol, C₆-OH, and C₈-OH appeared simultaneously at the beginning of the reaction. The concentrations continued to increase until a reaction time of 900 min in the order of butanol > C₆-OH > C₈-OH. The concentrations of butanol and C₆-OH began to decrease, possibly due to the side reaction of decarbonylation. When the Ni/alkaline site ratio was tuned by reducing the Ni loading amount, the porous BAP-0.25Ni catalyst exhibited obviously different behavior. The products of butanol, C₆-OH, and C₈-OH appeared in the order of butanol > C₆-OH ≈ C₈-OH at the beginning of the reaction. The concentrations of C₆-OH ≈ C₈-OH gradually grew and exceeded that of butanol ca. 1000 min, which cannot be explained by the sequential propagation reaction mechanism. The C₆+OH products were substantially higher than the Anderson–Schulz–Flory distribution.

Possible Pathway for Ethanol Upgrading. There are two viewpoints on the pathway of upgrading lower alcohols to higher alcohols. The first is the direct coupling pathway (Scheme S1A),^{15,53} and the second is the Guerbet pathway (Scheme S1B).^{54–57} According to the Guerbet pathway, the initial alcohols first dehydrogenate to acetaldehydes, then the acetaldehydes proceed via aldol condensation to produce alkenal, and finally the formed alkenal is hydrogenated to a saturated alcohol. The direct coupling route holds that two molecules of lower alcohols are directly coupled to form alcohols with double carbon numbers without the hydrogenation/dehydrogenation and aldol condensation of acetaldehyde.

To clarify the high selectivity toward C₆+OH in this work, the reaction pathway for ethanol coupling was investigated. First, the ethanol upgrading reaction was placed under different hydrogen pressures to verify whether the presence of hydrogen had an effect on the reaction results. As shown in Figure 12, ethanol conversion decreased with increasing initial hydrogen pressure. It would be a plausible explanation that the in situ formed hydrogen inhibited the ethanol dehydrogenation reaction, which is the “first step” of the Guerbet reaction. On the other hand, C₆+OH selectivity also decreased with increasing initial hydrogen pressure. This is because under

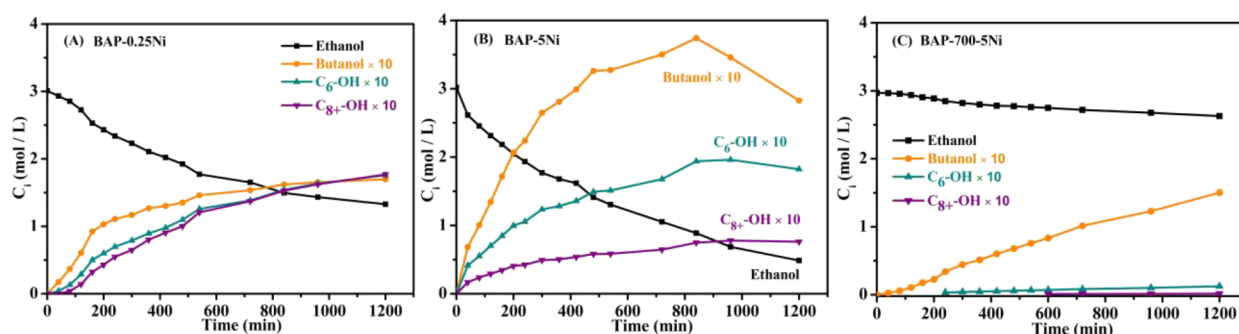


Figure 11. Ethanol upgrading to higher alcohols over (A) BAP-0.25Ni, (B) BAP-5Ni, and (C) BAP-700-5Ni in the 100 mL batch reactor. Conditions: 0.1 MPa N_2 , 200 °C, 20 h, 50 mL of toluene, 10 mL of ethanol, and 5 g of the catalyst.

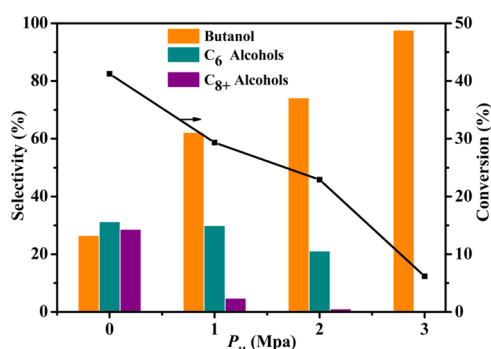


Figure 12. The effect of initial hydrogen pressure on reaction. Conditions: 200 °C, 24 h, 2.5 mL toluene, 0.5 mL ethanol, 0.25 g catalyst.

high hydrogen pressure, low-carbon aldehydes, such as acetaldehyde and butyraldehyde, are easier to hydrogenate to corresponding alcohols, cutting off the route further condensing to higher-carbon products. To gain more insights into the reaction pathway, the possible acetaldehyde intermediate was suddenly injected into the reactor at 120 min (Figure 13). As a

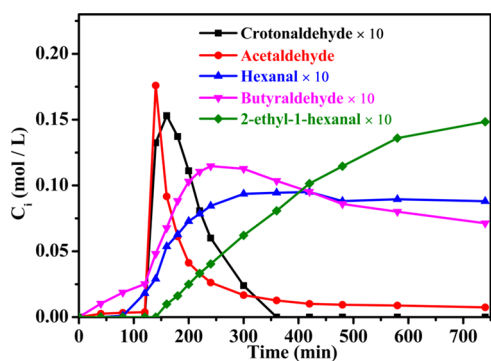


Figure 13. Ethanol upgrading to higher alcohols using BAP-0.25Ni. Aldehyde concentrations plotted as a function of time. Conditions: 0.1 MPa N_2 , 200 °C, 12 h, 50 mL of toluene, 10 mL of ethanol, 5 g of the catalyst.

response, crotonaldehyde, which is a key intermediate in the Guerbet reaction, immediately increased and then decreased as the reaction progressed. In addition, the rate of butyraldehyde and hexanal formation was also significantly accelerated. Therefore, in this work, upgrading ethanol to higher alcohols was more likely to follow the Guerbet pathway instead of the direct coupling pathway.

Key Factors for Upgrading Ethanol to Higher Alcohols.

A high ratio of C_{6+} -OH can be produced with a tuned Ni/BAP catalyst under suitable reaction conditions, as demonstrated in the catalytic performance test. This is due to the appropriate metallic/basic sites ratio and the “confinement effect” of the mesoporous structure. When there are few metallic sites (BAP-0.1Ni), the dehydrogenation ability of the catalyst is weak, and the concentration of aldehyde intermediate is low, slowing down the aldol reaction. Therefore, both the ethanol conversion and C_{6+} -OH selectivity are very low. With the increase of metallic sites (0.25–1 wt %), the processes of dehydrogenation and aldol reaction are immediately promoted, and ethanol conversion and C_{6+} -OH selectivity reach a high level. However, if there are excess metallic sites (HAP-5Ni), the original basic sites which facilitate the aldol reaction will be covered. Moreover, the decarbonylation will be accelerated and the alkanes increase, resulting in decreased selectivity of C_{6+} -OH.

Compared with the nonporous catalyst, the porous catalyst of BAP-0.25Ni exhibited remarkably high selectivity for C_{6+} -OH due to the “confinement effect” of the mesopore structure. A reaction network for ethanol upgrading was proposed according to reaction results and publications.^{24,55,56} As illustrated in Figure 14, in the case of a nonporous catalyst, ethanol molecules were dehydrogenated at Ni^0 active sites, producing acetaldehyde. Then, acetaldehyde molecules can be coupled to acetaldol through aldol reaction, followed by dehydration to crotonaldehyde at the base sites of BAP. Finally, crotonaldehyde is hydrogenated to butanol. In the process, the intermediates or butanol can freely leave the catalyst surface by diffusion. They can also return to the catalyst surface by adsorption to continue the reaction to give longer chain alcohols when they accumulate to a certain concentration. Thus, sequential chain propagation could be observed. However, in the case of the porous catalyst, ethanol molecules entered the pores of the catalyst and were then dehydrogenated to produce acetaldehyde at Ni^0 active sites in the pores. Acetaldehyde molecules cannot quickly leave the catalyst surface due to the “confinement effect”. The aldol reaction of acetaldehyde easily gave long chain aldehydes due to abundant basic sites in the vicinity. Then, the long chain aldehydes can be further hydrogenated to the corresponding alcohols.

Comparison with Reported Results. Our results have been compared with some other reported catalysts. Figure 15 shows that the catalysts in this work did achieve excellent C_{6+} -OH yield in the liquid–solid reaction system at a relatively low temperature lower than 200 °C. Other advantages are also

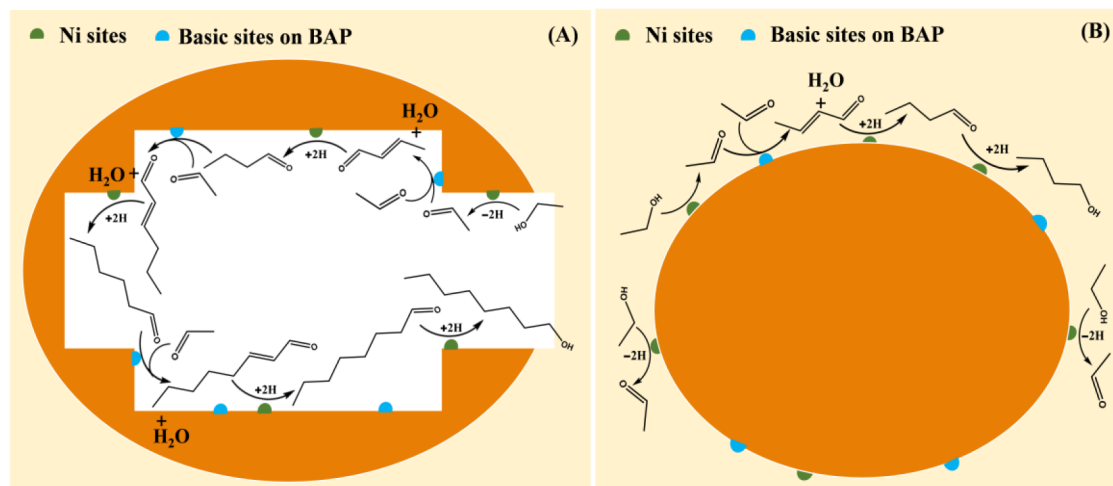


Figure 14. Reaction pathway on (A) porous BAP-Ni; (B) nonporous BAP-Ni catalyst.

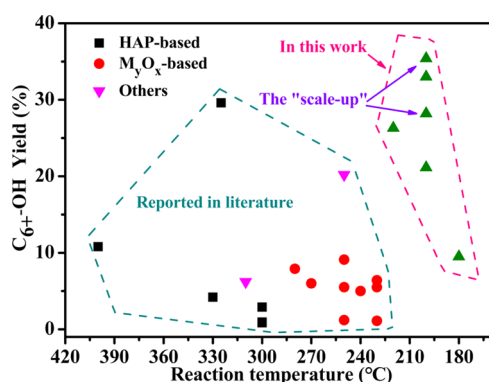


Figure 15. Yields of C_{6+} -OH versus reaction temperature for catalysts described in this work and in the literature.^{19–22,24,54,58–68} Abscissa value: Yields of C_{6+} -OH (%). Ordinate value: reaction temperature.

obvious: (i) no ligand complex; (ii) no noble metal; (iii) low Ni metal loading; (iv) utilization of natural waste; (v) no addition of nonrecyclable alkaline additives; and (vi) recyclable heterogeneous system catalyst.

Recycle and Regeneration of the Catalyst. BAP-0.25Ni was used to test the reusability of the catalyst. The results are shown in Figure 16. The catalytic activity and selectivity for C_{6+} -OH decreased significantly in the first three recycling tests. The ethanol conversion and C_{6+} -OH selectivity decreased from 33.3 to 8.6 and 63.5 to 40.9%, respectively. However,

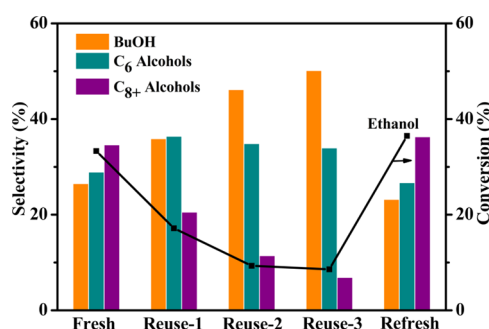


Figure 16. Reusability of the BAP-0.25Ni catalyst. Conditions: 0.1 MPa N_2 , 200 °C, 24 h, 2.5 mL of toluene, 0.5 mL of ethanol, and 0.25 g of the catalyst.

after reducing the catalyst at 470 °C, the activity of the catalyst was completely restored. The ethanol conversion and C_{6+} -OH selectivity reached 36.5 and 63.0%, respectively.

The comparison of the structure of the catalyst before and after use is shown in Figure 17. From the infrared results (Figure 17A), there was no significant change in the functional group of the BAP-0.25Ni catalyst. The XRD patterns show that slight mineralization occurred on BAP-0.25Ni used after three recycles (Figure 17B). The specific surface area also slightly decreased to 75 m^2/g , but the mesoporous structure was still maintained (Figure 17C). The Ni^0 content dropped from 21.2 to 10.2%, as shown by XPS (Figure 17D). Figure 18 shows that Ni was still well dispersed on the BAP without obvious aggregation phenomenon.

Combining the structural changes of the catalyst, the main reason for its deactivation was very likely that Ni^0 was oxidized. Ni^0 is responsible for ethanol dehydrogenation, so the activity of the catalyst decreases significantly after Ni^0 is oxidized, which reduces the concentration of the aldehyde intermediate. Therefore, the efficiency of the aldol reaction also decreases, resulting in a reduction in the selectivity of C_{6+} -OH, as was the result of the reaction on BAP-0.1Ni. The catalyst activity can be completely restored merely after reducing the Ni/BAP catalyst again.

CONCLUSIONS

A series of mesoporous bio-apatites with different pore diameters were obtained by calcining pork bones at different temperatures and then employed as carriers of Ni-based catalysts for upgrading ethanol to higher alcohols. Among the different catalysts tested, BAP-0.25Ni was demonstrated to be the most efficient catalyst. An ethanol conversion of 55.6% and a total alcohol selectivity of 90.3% with 22.6% *n*-butanol, 30.9% C_6 -OH, and 36.8% C_{8+} -OH could be achieved under the optimized reaction conditions. In particular, 67.7% C_{6+} -OH and 36.8% C_{8+} -OH selectivities were both the highest in the reported literature.

The results showed that the mesoporous structure played a vital role in promoting the condensation of lower alcohols. It could provide more basic active sites and increase the carbon chain through its “confinement effect” during the reaction. The differences in the activity and selectivity were also related to the metallic/basic sites in the fresh catalysts. A relatively low

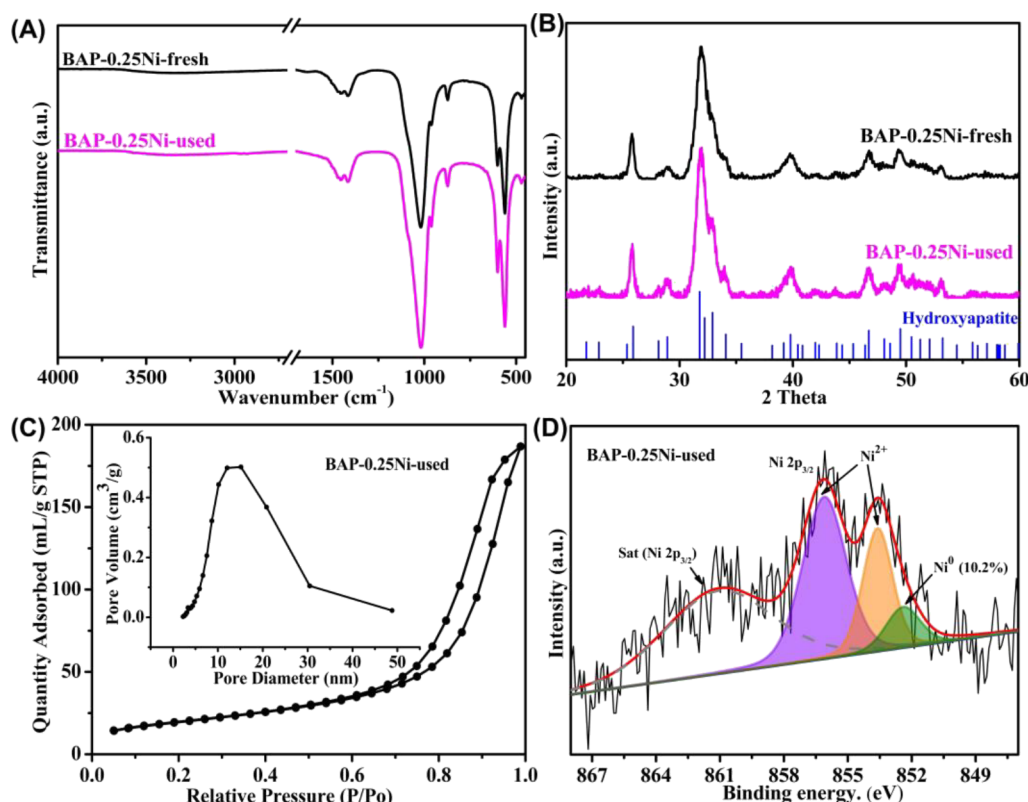


Figure 17. (A, B) FTIR and XRD of fresh BAP-0.25Ni and BAP-0.25Ni used four times; (C, D) BET and XPS of BAP-0.25Ni used four times.

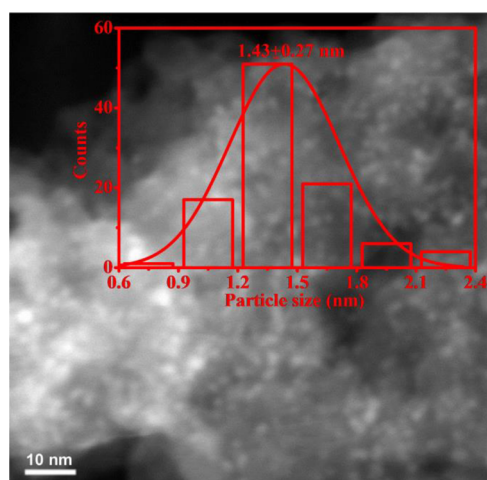


Figure 18. STEM images of refreshed BAP-0.25Ni.

Ni content was required to decrease alkane formation and favor aldol condensation to C_6+ -OH. In addition, Ni^0 was partially oxidized to Ni^{2+} due to the coproduced water, deactivating the catalyst. However, the activity could be restored by reducing again. With the rapid development of bioethanol, this more effective and environmentally friendly process for upgrading bioethanol to higher alcohol might have broader application prospects.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.1c07189>.

Additional catalyst characterization data, sample gas chromatography results, and schematic diagram of the ethanol upgrading path (PDF).

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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